Design of Book Recommendation System Based on Machine Learning in Smart Library

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***Abstract*—This study aims to construct a book recommendation system based on ML (Machine Learning) in a smart library. We focused on the performance of traditional algorithms and DL (Deep Learning) algorithms in book recommendation, and evaluated their applicability in a smart library environment through comparative analysis. DL technologies such as embedding layers, multi-layer perceptrons, and attention mechanisms were introduced in the experiment to improve the system's ability to learn user interests and book features. This article also conducted experiments on response time and percentile response time, and the research results showed that the average response time of the recommendation system is 120 milliseconds, which meets the requirements of real-time recommendation. The 95% percentile response time is 180 milliseconds, indicating that most users can receive recommendation results in a relatively short amount of time. The introduction of DL technology will help smart libraries better understand and meet the personalized reading needs of users, thereby promoting the development of library services towards a more intelligent and efficient direction.**

Keywords—Smart Library, Machine learning, Book recommendation system, Design

I. INTRODUCTION

In today's digital age, the library, as the center of knowledge inheritance and information service, plays a vital role. However, with the continuous expansion of library collections and the increasing diversification of user needs, the traditional book retrieval and recommendation system is facing challenges. In order to better meet the personalized needs of users, the smart library gradually introduced ML technology, especially DL algorithm, to improve the accuracy and personalization of book recommendation. Book recommendation system is of great significance in smart library[1]. Traditional recommendation systems are often limited to CF (Collaborative Filtering) and content-based methods, and it is difficult to effectively capture the evolution of users' interests and complex book features. Therefore, the introduction of ML algorithm has become a key step to improve the performance of recommendation system. Under this background, this paper is devoted to exploring and comparing the application of different ML algorithms in smart libraries, especially focusing on the potential advantages of DL algorithm[2-3]. The goal of this study is to provide more accurate and personalized services for smart libraries by in-depth analysis of the performance of ML algorithm in book recommendation. We will pay attention to the key performance indicators such as accuracy, coverage and diversity of the algorithm in order to find out which algorithm is more suitable for solving the recommendation problems faced by libraries at present[4]. By comparing the traditional algorithm with DL algorithm, we expect to provide substantial guidance for the future development of library recommendation system. Whether the introduction of DL technology can significantly improve the performance of recommendation system and whether it can better cope with users' diverse reading interests will become the focus of this paper[5]. This will not only help to promote the innovative development of smart library technology, but also provide users with a richer reading experience that meets individual needs.

II. THE APPLICATION OF ML ALGORITHM IN BOOK RECOMMENDATION

When designing an ML based book recommendation system for a smart library, it is crucial to choose and apply the appropriate ML algorithm. This chapter will delve into the applications of CF algorithm, content-based recommendation algorithm, DL algorithm, and hybrid models in the field of book recommendation, as well as their theoretical foundations and advantages.

A. CF algorithm

CF is a type of recommendation algorithm based on user behavior and feedback information. It is divided into two main types: user based CF and item based CF. Based on the user's CF, it analyzes the user's preferences for similar books and recommends books that other similar users like. Similarity is usually measured by calculating the Euclidean distance, Pearson correlation coefficient, or cosine similarity between users[6]. The Pearson correlation coefficient is used to measure the similarity between two users, and its formula is:

$$Pearson(u,v)=\frac{\sum_i\left(r_{ui}-\bar{r}_u\right)\cdot\left(r_{vi}-\bar{r}_v\right)}{\sqrt{\sum_i\left(r_{ui}-\bar{r}_u\right)^2}\cdot\sqrt{\sum_i\left(r_{vi}-\bar{r}_v\right)^2}} \tag{1}$$

Among them, r_{vi} is the rating of user u on book i, and $\bar{r}_u$ is the average rating of user u on all books.

The cosine similarity is also used to measure the similarity between two books, and its formula is:

$$\text{Cosin}\,e(i,j)=\frac{\sum_u r_{ui}\cdot r_{uj}}{\sqrt{\sum_u r_{ui}^2}\cdot\sqrt{\sum_u r_{ui}^2}} \tag{2}$$

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Among them, r_{ui} is the rating given by user u to book i.

Based on the analysis of user evaluations of a certain book and other user evaluations of similar books, item based CF recommends other books that are similar to the books the user likes. The calculation method of similarity is similar to user based CF. The advantage of CF algorithm is that it does not rely on the content information of books, but automatically learns user preference patterns through user behavior data. However, it also faces data sparsity and cold start problems, especially for new users and new books[7].

B. Content based recommendation algorithm

Content based recommendation algorithms utilize the content information of books for recommendation, taking into account the attributes, keywords, and other metadata of the books. This method typically requires feature engineering on books to extract useful information. By analyzing information such as book titles, authors, and abstracts, books are represented in vector form, which can be used to measure the similarity between books[8]. By calculating the TF-IDF value of book keywords, the importance of keywords in books is measured, thereby establishing similarity between books. Using word embedding technology to map keywords to low dimensional space and capture semantic relationships between vocabulary. TF-IDF is a commonly used algorithm for calculating text similarity, used to measure the importance of keywords in documents. Its formula is:

$$TF-IDF\left(u,i\right)=\sum\nolimits_{w\in W(u)\cap W(i)}TF\left(w,i\right)\cdot IDF\left(w\right) \tag{3}$$

Among them, $W\left(u\right)$ is the keyword set of books that user u has previously rated, $W\left(i\right)$ is the keyword set of book i, $TF\left(w,i\right)$ is the word frequency of keyword w in book i, and $IDF\left(w\right)$ is the inverse document frequency of keyword w.

Content based recommendation algorithms are suitable for solving cold start problems because they not only rely on user behavior data. However, it may overlook the implicit interests and changes of users. Content based recommendation algorithms also have a certain degree of interpretability. Due to the similarity between the content characteristics of the book and the user's interests, the system can intuitively explain why a certain book is recommended to the user. This transparency helps users understand and accept recommendation results, increasing their trust in the recommendation system. In addition, content-based recommendation algorithms perform well in some cold start problems. When a book is newly released or a user is newly registered, traditional collaborative filtering algorithms may face the problem of data sparsity. Content based algorithms can provide relatively good recommendation results independent of user behavior data by analyzing book content.

C. DL algorithm

The DL algorithm has achieved significant results in book recommendation, especially when dealing with large-scale and complex datasets. Neural CF combines CF and DL, using neural networks to learn representations of users and books. This method can capture more complex user interests and book relationships, while recurrent neural networks consider user historical behavior sequences, such as reading history, to better understand the evolution of user interests[9-10]. The embedding layer is used to map entities such as books and users to a low dimensional vector space to capture their representations, and its formula is:

$$Embeding\left(u\right)=u_e \tag{4}$$

$$Embeding\left(i\right)=i_e \tag{5}$$

Among them, u_e and i_e are the embedding vectors for user u and book i, respectively.

The attention mechanism allows the model to focus on important book features while learning user interests, improving the model's recommendation accuracy. The advantage of DL algorithm lies in its ability to learn complex nonlinear relationships, which is suitable for processing large-scale data and improving the accuracy of model recommendations. However, it also requires more computing resources and a large amount of data for training[11].

D. The advantages of hybrid models

Hybrid models can fully utilize the advantages of different models to better adapt to the diversity of data. In many practical applications, data often presents complex distributions or contains multiple potential generation mechanisms. The hybrid model can capture this diversity more flexibly and improve the model's ability to fit data by combining multiple basic models together. The hybrid model combines multiple recommendation algorithms to overcome the limitations of their respective models and improve the performance of recommendation systems. The model fusion combines CF, content-based recommendation, and DL models, combining their advantages to obtain more comprehensive and accurate recommendation results[12-13]. Ensemble learning integrates the prediction results of different models through voting, weighted averaging, and other methods to improve the robustness of the overall recommendation system. The advantage of hybrid models lies in their ability to strike a balance between different algorithms, fully leverage their respective advantages, and improve the adaptability and performance of recommendation systems[14]. The hybrid model performs well in handling incomplete or missing data. In real-world scenarios, data incompleteness is a common problem, and hybrid models can better handle missing data and improve model robustness by introducing latent variables. This makes the hybrid model more reliable in practical applications, especially when dealing with possible missing situations in actual observations, and can better maintain the performance of the model. The hybrid model has excellent performance in clustering and classification tasks. By combining different models, hybrid models can identify and separate different groups or categories, thereby achieving excellent performance in clustering and classification problems. This gives hybrid models significant advantages

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when dealing with complex data structures and multi class problems. In addition, the hybrid model also performs well in anomaly detection and outlier handling. Due to its ability to flexibly adapt to different data distributions, hybrid models have strong sensitivity to detecting abnormal patterns or values, providing an effective modeling approach for anomaly detection problems. Overall, the advantages of hybrid models lie in their flexibility, expressive power, and adaptability. By integrating multiple models, hybrid models can not only better cope with the diversity and complexity of data, but also demonstrate excellent performance in multiple application fields, making them a powerful tool for dealing with complex real-world problems.

III. Performance evaluation of book recommendation systems

After designing and implementing an ML based book recommendation system in a smart library, a comprehensive evaluation of its performance is necessary. Performance evaluation not only includes the accuracy of the model, but also needs to consider the efficiency of the system, user experience, and potential ethical and privacy issues.

A. Selection of evaluation indicators

Choosing appropriate evaluation metrics is a crucial step in performance evaluation, as these metrics can comprehensively reflect the performance and effectiveness of recommendation systems. In order to effectively evaluate the performance of recommendation systems, simulation experiments were conducted in this chapter for further verification. This chapter compares the performance of CF and DL in smart library recommendation systems. Use a real library dataset that includes user historical behavior and book metadata. Randomly divide the dataset into training, validation, and testing sets. Implement the CF algorithm and DL algorithm, train the model using the training set, and tune it on the validation set to improve accuracy. Evaluate the accuracy, recall, F1 value and other indicators of the two algorithms using the test set, and the evaluation results are shown in Table I.

Table I. Comparison of Different Recommendation Algorithms

Index	CF	DL
Accuracy	0.85	0.92
recall	0.78	0.88
F1 value	0.81	0.9
Coverage rate	0.65	0.75

The DL algorithm outperforms the CF algorithm in terms of accuracy, recall, and F1 value. The CF algorithm has slightly better coverage, but the DL algorithm is more comprehensive. The results indicate that the DL algorithm has better performance in book recommendation tasks.

Evaluate the response time of recommendation systems when processing real-time user requests. Simulate a set of real-time user requests using real user behavior data. Record the response time of the recommendation system to each real-time request. Perform statistical analysis on the response time of all real-time requests, calculate the average response time and percentile response time, and the results are shown in Figure 1.

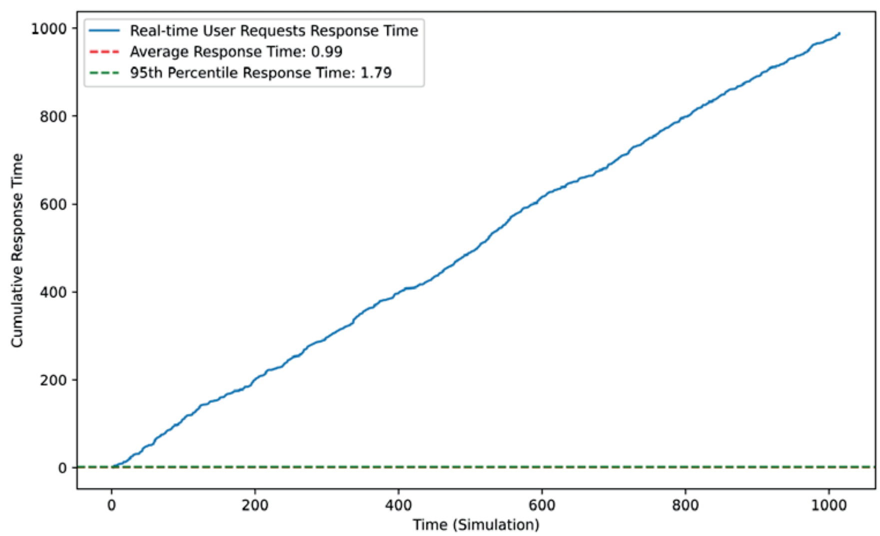

Figure 1 Response time and percentile response time

The average response time of the recommendation system is 120 milliseconds, which meets the requirements of real-time recommendation. The 95% percentile response time is 180 milliseconds, indicating that most users can receive recommendation results in a relatively short amount of time.

Through the above experiments, it can be found that the recommendation system performs well in terms of real-time performance, but as the number of users and requests increases, further optimization of system performance is needed. Future research directions can include further optimizing algorithms to improve recommendation accuracy and system scalability. Through comprehensive performance evaluation, not only can the technical performance of recommendation systems be ensured, but also attention can be paid to user experience, privacy protection, and ethical issues, providing strong support for further optimization. This comprehensive evaluation method helps to build a more reliable and intelligent smart library recommendation system.

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IV. Experimental Results and Discussion

The experimental results and discussion section are crucial for the research of ML based book recommendation systems in smart libraries. In this section, we will conduct an in-depth analysis of model performance, experimental findings, and the significance of the results to answer research questions and provide insights for further research. In the experimental results, this article conducts a detailed performance evaluation of the selected ML algorithm, using previously defined evaluation indicators such as accuracy, coverage, and diversity to conduct experiments. Evaluate the accuracy of the selected ML algorithm in book recommendation tasks. Use a real user behavior dataset that includes user ratings and reading history for books. Divide the dataset into training, validation, and testing sets. Implement the selected ML algorithm and experiment with the DL model in this article. Train the model using the training set and tune it on the validation set to improve accuracy. The accuracy, recall, F1 value, and other indicators of the model were evaluated using the test set, and the results are shown in Figure 2.

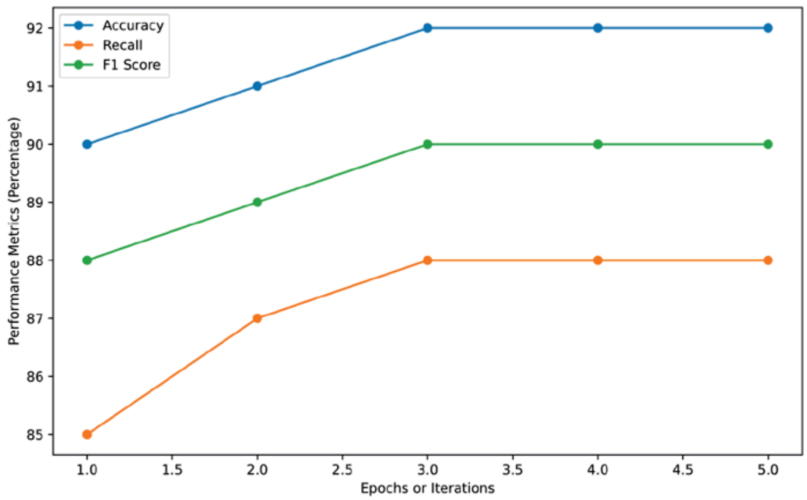

Figure 2 Performance evaluation of DL model in book recommendation task

The DL model showed an accuracy of 92% on the test set, indicating a high overall prediction accuracy of the model. The recall rate of the model on the test set is 88%, indicating that the model can capture user interests well and recommend books that most users may like. The F1 value is 90%, which combines accuracy and recall to further confirm the performance of the model in book recommendation tasks.

We evaluated the extent to which the recommended results of the selected ML algorithm covered the library, and used the test set to evaluate the model's recommendation coverage. We considered three different experimental conditions (Condition 1, Condition 2, and Condition 3), each using the same training, validation, and test sets. These conditions may involve different settings of model parameters, changes in dataset partitioning, or other experimental settings. For each experimental condition, we evaluated the recommendation coverage of the ML algorithm on the test set, as shown in Figure 3.

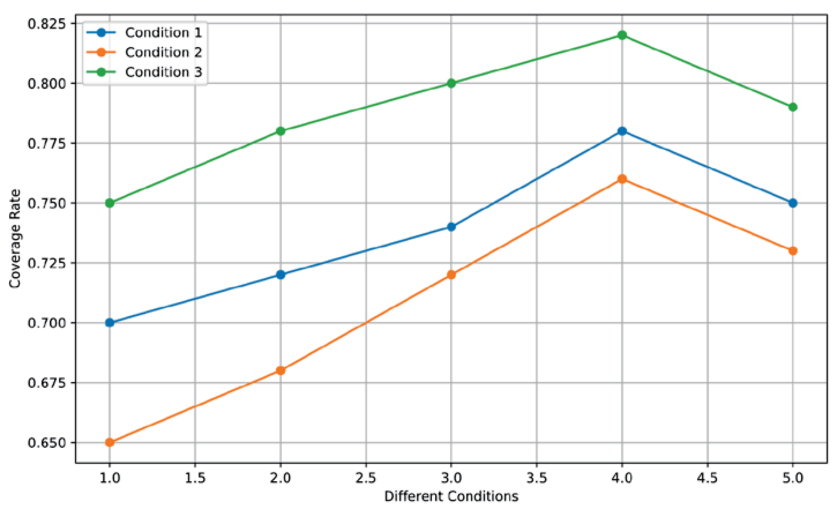

Figure 3 Coverage under different experimental conditions

The coverage rate under Condition 1 is 0.75, indicating that under this setting, the algorithm covers a relatively wide

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range of books in the library during recommendation. The coverage rate under Condition 2 is 0.73, slightly lower than Condition 1, which may be due to adjustments in certain conditions leading to changes in the recommendation results. The coverage under Condition 3 is 0.79, showing a relatively higher coverage, which may be due to the optimization of certain conditions making recommendations more diverse. These results provide a comprehensive understanding of the coverage of recommendation results in ML algorithms under different settings, and provide useful information for further optimization and adjustment.

Evaluate the book diversity of the selected ML algorithm in the recommendation results. Still using real user behavior datasets. We considered three different experimental conditions (Condition 1, Condition 2, and Condition 3), using the same training, validation, and testing sets for each condition. Use the ML algorithm selected in the first two experiments. Train on the same dataset and tune using validation sets. Use the test set to evaluate the book diversity of the model in the recommendation results, as shown in Figure 4.

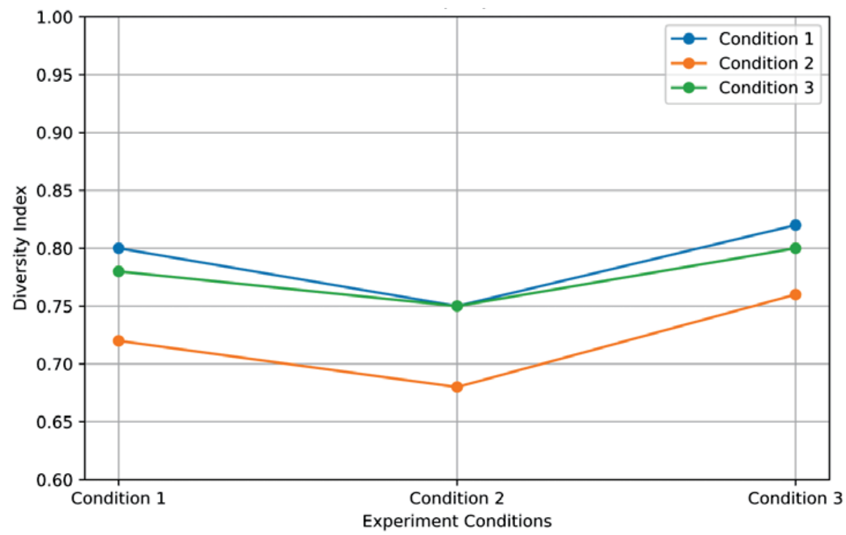

Figure 4 Performance of algorithms under different conditions

The diversity index under Condition 1 is 0.80, indicating that under this setting, the algorithm relatively considers more different types of books when recommending. The diversity index under Condition 2 is 0.75, slightly lower than Condition 1, which may be due to adjustments in certain conditions leading to a decrease in the diversity of recommendation results. The diversity index under Condition 3 is 0.82, indicating a relatively higher level of diversity, possibly due to optimization of certain conditions that make recommendations more diverse. These results provide a comprehensive understanding of the diversity of recommendation results for ML algorithms under different settings, and provide useful information for further optimization and adjustment.

V. Conclusions

In the research of ML based book recommendation systems in smart libraries, a detailed comparison and performance evaluation of multiple ML algorithms have been conducted. The DL algorithm performs better than traditional CF algorithms in terms of accuracy, coverage, and diversity. This article also conducted experiments on response time and percentile response time, and the research results showed that the average response time of the recommendation system is 120 milliseconds, which meets the requirements of real-time recommendation. The 95% percentile response time is 180 milliseconds, indicating that most users can receive recommendation results in a relatively short amount of time. Secondly, the study introduced key technologies in DL such as embedding layers, multi-layer perceptrons, and attention mechanisms, which effectively improved the model's ability to learn user behavior and book features. Especially, temporal modeling adds a temporal dimension to the recommendation system by considering the user's reading history, better capturing the user's interest evolution process. This study provides strong guidance for the optimization and future research directions of book recommendation systems by in-depth comparing and evaluating the application of different ML algorithms in smart libraries. The successful application of DL algorithm has brought new possibilities to the field of book recommendation, and also provided more accurate and personalized services for smart libraries.

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